

Foundations of Language Processing: From Morphology to Semantics

Exploring how computers understand human
language

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| Language & Knowledge Transmission

Language is the primary system humans use to encode concepts and transmit knowledge from one mind to another.

- ✓ **Encoding:** Turning an abstract thought into words.
- ✓ **Transmission:** Sharing those words so someone else can decode the thought.

BASIC EXAMPLE

Thought: You feel cold.

Encoding: You say, "*Please close the window.*"

Transmission: Your friend hears it and understands the action needed to fix your coldness.

| Ambiguity and Context

Language is inherently messy. **Ambiguity** happens when a word, phrase, or sentence has more than one meaning.

We rely heavily on **Context** (the surrounding words or situation) to figure out the correct meaning.

EXAMPLE OF AMBIGUITY

Sentence: *"I saw the man with the binoculars."*

Meaning A: I used binoculars to see the man.

Meaning B: I saw a man who was holding binoculars.

Without context, a computer cannot know which is true!

| Morphology: Tokenization

Morphology is the study of how words are formed. The first step for a computer to process language is breaking it down.

Tokenization is the process of chopping a long string of text into smaller pieces, called "tokens" (usually words or punctuation).

TOKENIZATION EXAMPLE

Input Sentence: "Hello, world! NLP is fun."

Tokens: ["Hello", ",", "world", "!", "NLP", "is", "fun", "."]

Stemming vs. Lemmatization

Both techniques aim to reduce a word to its base form, but they do it differently.

STEMMING (THE "CHOP" METHOD)

Uses simple rules to chop off the ends of words. It is fast but can make mistakes.

Example:

"Studying" → "Study"

"Studies" → "Studi" (Not a real word!)

LEMMATIZATION (THE DICTIONARY METHOD)

Looks up the word in a vocabulary to find the true root (the "lemma"). Slower, but accurate.

Example:

"Better" → "Good"

"Studies" → "Study"

| Part-of-Speech (POS) Tagging

Assigning a grammatical category (Noun, Verb, Adjective, etc.) to each word in a sentence.



Rule-Based Taggers

Uses grammar rules. E.g., "If a word ends in '-ed', tag it as a Past Tense Verb."



Machine Learning Taggers

Learns from massive amounts of pre-tagged text to guess the tag based on word probability.

EXAMPLE

Sentence: "The quick dog barks."

Output: The [Det] quick [Adj] dog [Noun] barks [Verb].

| Lexical Semantics & WordNet

Lexical Semantics is the study of word meanings and how words relate to each other.

WordNet is a massive, publicly available lexical database for English. It acts like a super-thesaurus for computers, grouping words together based on their meanings.

WORDNET CONCEPT EXAMPLE

If you search for "Car" in WordNet, it tells the computer:

1. It is synonymous with "Automobile".
2. It is a type of "Vehicle".
3. It has parts like "Wheels" and "Engine".

Word Relationships

Synonymy

Words that have the exact or very similar meaning.

Example:
"Large" and "Big"

Hyponymy

An "is-a-type-of" relationship.
One word is a subclass of another.

Example:
"Dog" is a hyponym of "Animal"

Polysemy

When a single word has multiple related meanings.

Example:
"Mouse" (Animal) vs. "Mouse" (Computer device)

| Word Sense Disambiguation (WSD)

Because of Polysemy (multiple meanings), computers must figure out exactly *which* meaning of a word is being used in a sentence. This is called Word Sense Disambiguation.

CONTEXT 1

*"I need to deposit a check at the **bank**."*

WSD Output: Financial Institution

CONTEXT 2

*"We pitched our tent on the river **bank**."*

WSD Output: Sloping land next to water

Comprehensive Guide to Text Preprocessing

Mastering the essential steps to clean and standardize unstructured text data for robust Machine Learning models.

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01. Introduction

Data cleaning is often more critical than model building. In NLP, preprocessing is the foundation of success.

Why Preprocess?

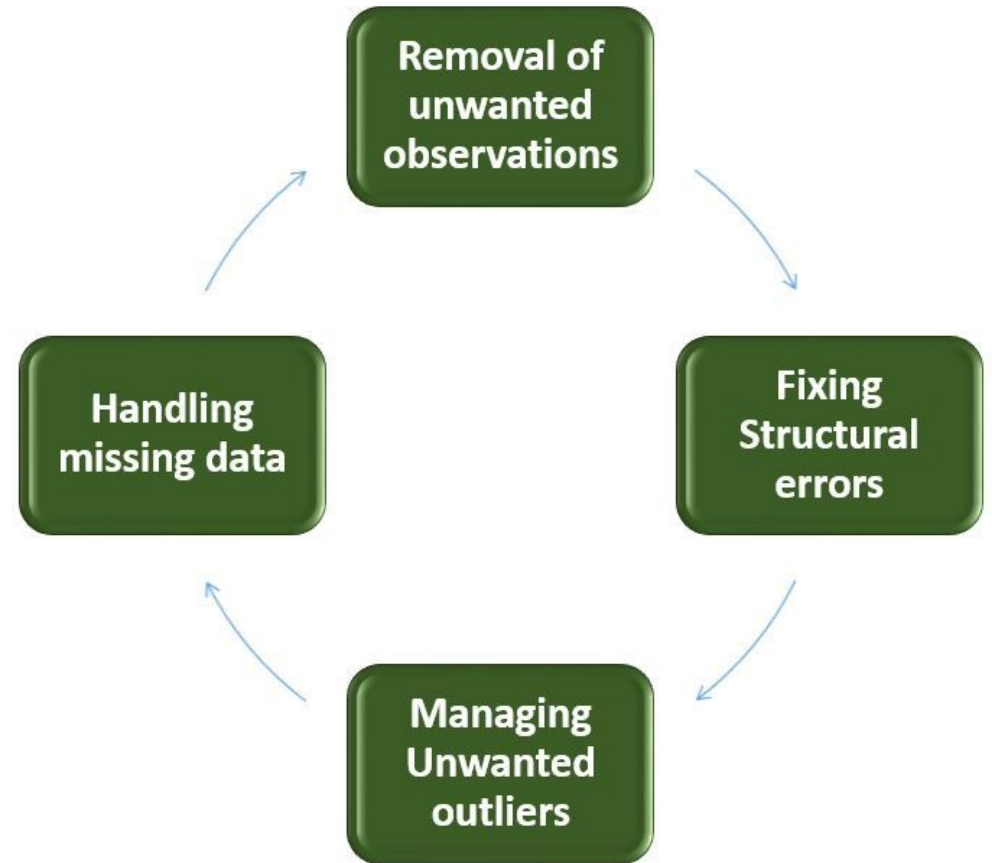
Cleaning unstructured text is vital because noise can confuse algorithms and degrade performance.

Objective: Understand various preprocessing steps and their implementation using code examples.

Note: Decisions must be use-case specific. E.g., sentiment analysis often requires keeping emojis while others may not.

Steps:

1. Removal of Unwanted Observations: Getting rid of text that offers no analytical value. Example: Stripping out HTML Tags
2. Fixing Structural Errors: Standardizing the format and normalizing the text so the computer reads it consistently. Example: Correcting character encoding issues (converting symbols to standard UTF-8), converting all text to lowercase, fixing typos.
3. Managing Unwanted Outliers: Filtering out text anomalies that could skew your analysis or break your model. Example: Removing documents that are written in the wrong language.
4. Handling Missing Data: Addressing blank, NULL or empty text fields. Example: Dropping rows where the text entry is completely blank.



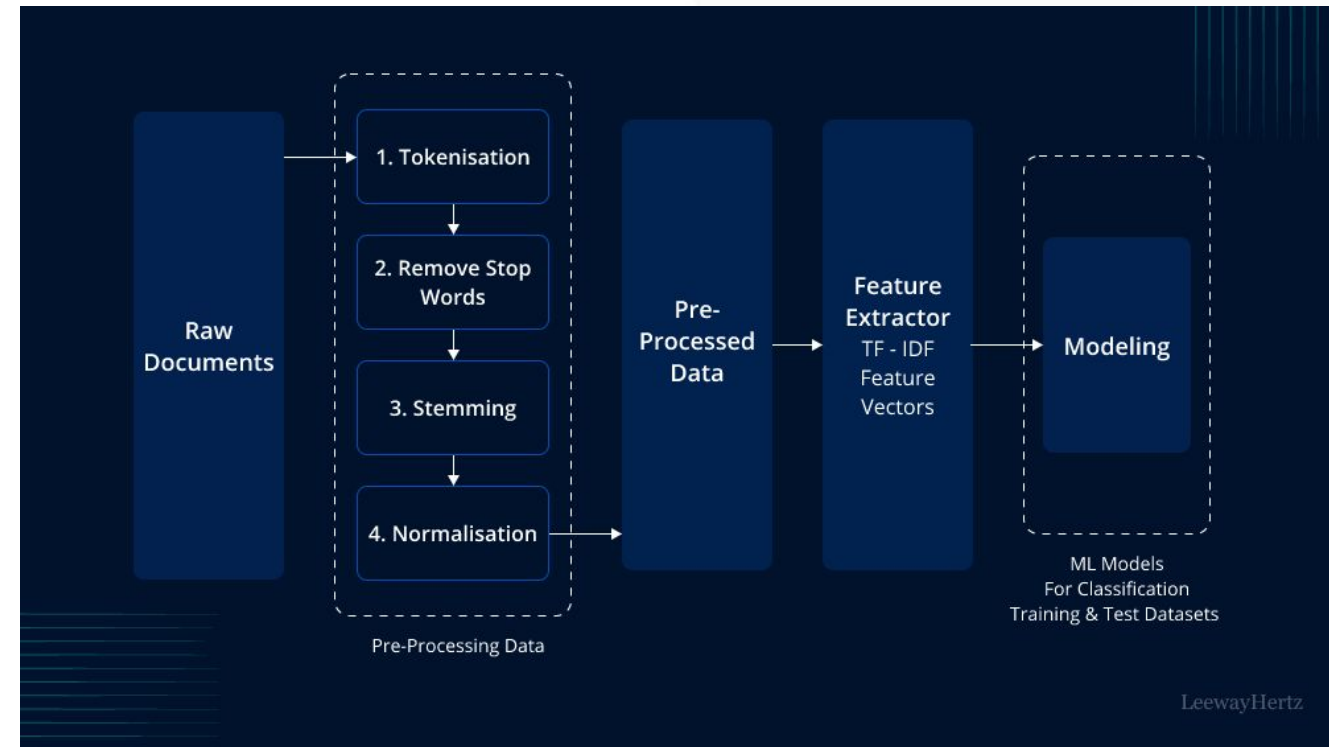
01. Lower Casing

Standardizes text so that different casings of the same word are treated identically.

- ✓ Helps frequency-based techniques (TF-IDF).
- ✓ Reduces vocabulary duplication.
- ✓ Caution: Avoid in POS tagging where casing identifies nouns.

EXAMPLE OUTPUT

"TEXT", "Text" → "text"



02. Removal of Punctuations

Theory

Removes symbols to treat base words correctly.

Standardization ensures 'hurray' and 'hurray!' are counted as the same entity.

Use Python's `string.punctuation` as a starting point, but customize based on requirements.

Example

Input: "@sprintcare I did."

Process: Remove @ and .

Output: "sprintcare I did"

03. Removal of Stopwords

- ✓ Stopwords are high-frequency words (e.g., 'the', 'is', 'a') that often carry little semantic meaning.
- ✓ Removing them reduces the dimensionality of the feature space.
- ✓ **Language-Specific:** Lists are available for most languages via libraries like NLTK.
- ✓ **Context Matters:** Do not remove for Part-of-Speech (POS) tagging or dependency parsing.

COMMON ENGLISH STOPWORDS

i, me, my, we, our, you, he, she, it, they, what, which, who, am, is, are, was, were, been...

| 04 & 05. Frequent and Rare Words



Frequent Words

Domain-specific terms that appear too often to be discriminative. Removing them is similar to using Stopwords but tailored to your specific corpus.



Rare Words

Words appearing only once or twice (noise/typs). Removing these helps prevent model overfitting on non-generalisable features.

06 & 07. Stemming vs Lemmatization

Stemming

Chops off suffixes (e.g., 'ing', 'ed') to find the base. Fast but crude.

Example: "running" → "run"

Risk: "private" → "privat"

Lemmatization

Uses a dictionary to find the linguistic root (lemma).
Slower but more accurate.

Example: "stripes" → "stripe" (noun) or "strip" (verb).

08-11. Emoji & Emoticon Handling



Removal

Standard clean-up to focus strictly on text characters. Useful for formal documents.



Conversion

Converting characters like :-) or 🔥 into words like "Happy" or "fire".
Essential for Sentiment Analysis.

EMOJI TO WORD CONVERSION

Input: "game is on 🔥" → Output: "game is on fire"

Input: "Hello :-)" → Output: "Hello Happy_face_smiley"

12 & 13. URL and HTML Tags

- ✓ **URL Removal:** Essential for social media data (Twitter/Reddit) where links don't add semantic value to text analysis.
- ✓ **HTML Tag Removal:** Critical when scraping web data. Tags like `<div>` or `<h1>` are noise.
- ✓ **Tools:** Regular expressions or libraries like BeautifulSoup for elegant extraction.

Input: `<h1>H2O AutoML</h1>` → **Output:** "H2O AutoML"

| 14 & 15. Chat Slang & Spelling

Chat Words

Expands abbreviations to full linguistic forms for better processing.

ASAP → As Soon As Possible

BRB → Be Right Back

Spelling Correction

Fixes typos using probabilistic models or dictionaries (e.g., pypellchecker).

"speling" → spelling

"readin" → reading

Questions?

Preprocessing is iterative. Always choose your steps based on the final machine learning objective.

References

1. [Getting started with Text Preprocessing](#)
2. [Difference between Data Cleaning and Data Processing - GeeksforGeeks](#)
3. [Natural Language Processing: A comprehensive overview](#)